

Risk Didn't Disappear. *Vulnerability Changed.*

Deaths relative to population fell sharply. Flood exposure persisted.
Consistent with adaptation—not proof of its causal contribution.

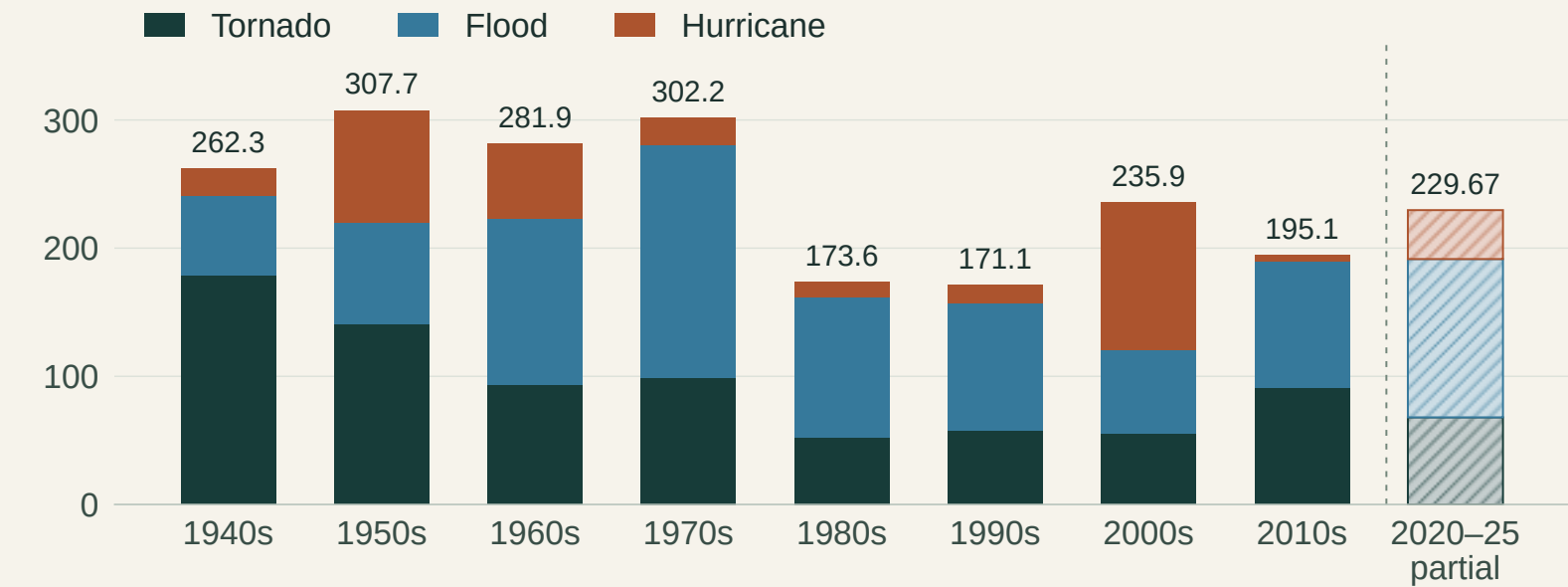
Caleb Katz

1.91 → 0.61 Combined annual deaths per million residents
1940s → 2010s

≈68% lower Tornado + Flood + Hurricane
Eight complete decades · NWS & Census [1, 2]

01 / The annual toll

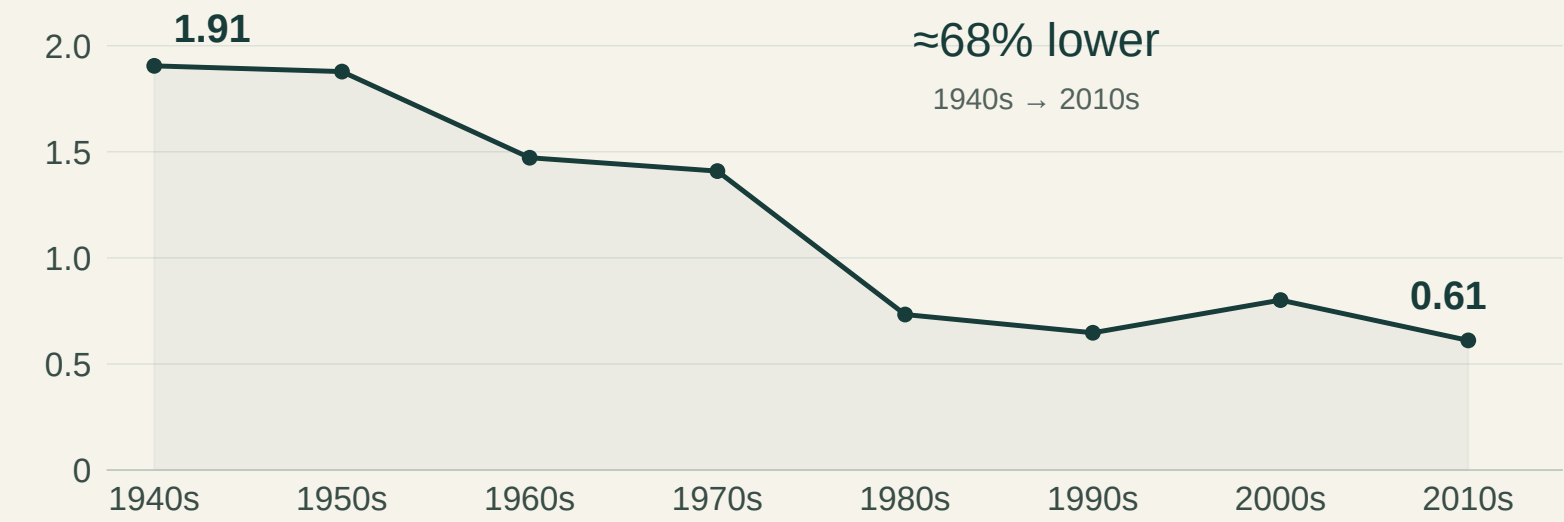
Average annual deaths · Tornado + Flood + Hurricane



NWS average annual reported deaths, 1940–2025. Three categories are stacked; totals are labeled. Hatched 2020–25 covers six years, with 2025 preliminary—not a complete decade. [1]

02 / Population changes the story

Combined annual deaths per million U.S. residents



Deaths per million residents per year, 1940s–2010s. Average deaths ÷ average population × 1,000,000. A descriptive statistic; no causal-model probability. Complete decades only. [1, 2]

Raw annual mortality fell about 26%, while average population grew from **137.7 million to 319.3 million**. Both trends matter; neither identifies the cause of changing mortality.

03 / MIGRATION & FLOOD EXPOSURE

It depends how closely you look.

Qin and Voorheis (2026) link flood risk to Census survey and administrative data. The exposed-household count rose; the exposed share was nearly flat. Coarse geography can change the migration story. [3]

Measure	Finding
Flood-exposed households	~11 million in 1999 → >15 million in 2023
Share of households exposed	Approximately 14.2%–14.4%
Census-block migration	Increasing net migration toward floodplain blocks
Parcel-level migration	Essentially no net migration toward flood-risk parcels

1999–2023 · Expansive FEMA parcel definition: A, V, or 500-year floodplains. Approximate figures remain approximate. A research working paper, not an official Census policy position.

Growth ≠ migration. ~11 million → >15 million exposed households is not evidence that migration added four million. Household formation, births, and international immigration differ from moves within the United States.

WHAT ECONOMICS ADDS

Risk = hazard × exposure × vulnerability

Conceptual framework, not an estimated equation. [5]

Forecasting, warnings, safer buildings, drainage, evacuation, emergency response, income, technology, and behavior can reduce vulnerability. Their roles are mechanisms to test, not effects identified by these charts.

People trade safety against jobs, housing costs, and amenities. Ask what protection costs, who pays, how insurance affects incentives, when public goods matter, and who lacks access to capital and reliable energy.

A DIFFERENT MORTALITY MEASURE

These numbers are not measuring the same thing.

Burke et al. estimate ambient-temperature effects using data from 30 countries: approximately **5–12% of all deaths across countries in their sample**. Cold contributes more than heat in the contexts studied; local adaptation does not eliminate exposure. [4]

NWS event counts

Reported fatalities assigned to weather-hazard categories.

Burke temperature attribution

Statistical estimates relating overall mortality to ambient temperature; many deaths need not be coded as heat or cold.

Do not add these estimates or compare them as competing totals. Higher income is not uniformly protective, and many proposed interventions lack strong causal evidence. [4]

Intervention test: cooling access has purchase, electricity, maintenance, and grid-reliability costs. Evaluate actual health effects and alternative uses of the budget—not simply the number of units supplied. Cost-effectiveness cannot be asserted without costs and credible effect estimates.

But wait—does declining mortality prove adaptation?

No. Hazard severity, event location, settlement, reporting, classification, population composition, and rare catastrophes also changed. NWS “Hurricane” is not every tropical-cyclone death: related flooding and tornadoes may be classified separately. Identifying adaptation’s contribution needs hazard/exposure controls and evidence isolating particular protections.

The evidence: mortality per resident in these three reported categories is much lower. **The limit:** it does not identify how much adaptation caused. Continued flood exposure can coexist with lower mortality; future protection is not free or automatic.

Sources & method

[1] NWS / NOAA, historical fatality series 1940–2025 and category definitions. [weather.gov/hazstat](https://www.weather.gov/hazstat).

[2] U.S. Census Bureau historical/intercensal population series (1900–1999, 2000–2010, 2010–2020). Decade means are used as a descriptive national denominator; definitions and estimate vintages vary across the long series. Endpoint decline = $1 - (195.1 / 319,333,559) / (262.3 / 137,686,400) \approx 67.93\%$. Not age/exposure adjusted.

[3] Suvy Qin & John Voorheis (2026), *Flood Risk, Insurance, and Housing in the United States*, Census CES-26-37, Section 3.1, Figures 2–3, pp. 9–11.

[4] Burke et al. (2025), *Understanding and Addressing Temperature Impacts on Mortality*, NBER WP 34313. doi.org/10.3386/w34313. [5] IPCC (2012), SREX.



Read the complete story

Charts, CSVs, definitions,
and direct source links

[calebjkatz.github.io/
econ238-portfolio/homework-4/](https://calebjkatz.github.io/econ238-portfolio/homework-4/)